

LAB EXPERIMENTS

Name : Vedvyass M

Reg no : 192324267

Experiment – 13 :

```
import pandas as pd
```

```
import numpy as np
```

```
# Create the DataFrame
```

```
data = {  
    'ord_no': [70001, np.nan, 70002, 70004, np.nan, 70005, np.nan, 70010,  
70003, 70012, np.nan, 70011],  
    'purch_amt': [150.50, 270.65, 65.26, 110.50, 948.50, 200.60, 5760.00,  
1983.43, 2480.40, 250.45, np.nan, np.nan],  
    'ord_date': ['2012-10-05', '2012-09-10', np.nan, '2012-08-17', '2012-09-10',  
                '2012-09-09', '2012-09-10', '2012-10-10',  
                '2012-06-27', '2012-08-17', '2012-04-25', np.nan],  
    'customer_id': [3002, 3001, 3001, 3003, 3002, 3001, 3004, 3003, 3002, 3001,  
3001, np.nan],  
    'salesman_id': [5002, 5003, 5001, np.nan, 5002, 5001, np.nan, 5003, 5002,  
5003, np.nan, np.nan]  
}
```

```
df = pd.DataFrame(data)
```

```
print("Original DataFrame:")
```

```
print(df)
```

```
print("\nMissing Values (True/False):")
```

```
print(df.isnull())
```

```
ord_no  purch_amt  ord_date  customer_id  salesman_id
0    70001.0    150.50  2012-10-05    3002.0    5002.0
1      NaN    270.65  2012-09-10    3001.0    5003.0
2    70002.0    65.26      NaN    3001.0    5001.0
3    70004.0    110.50  2012-08-17    3003.0      NaN
4      NaN    948.50  2012-09-10    3002.0    5002.0
5    70005.0    200.60  2012-09-09    3001.0    5001.0
6      NaN   5760.00  2012-09-10    3004.0      NaN
7    70010.0   1983.43  2012-10-10    3003.0    5003.0
8    70003.0   2480.40  2012-06-27    3002.0    5002.0
9    70012.0    250.45  2012-08-17    3001.0    5003.0
10     NaN      NaN  2012-04-25    3001.0      NaN
11   70011.0      NaN      NaN      NaN      NaN

Missing Values (True/False):
ord_no  purch_amt  ord_date  customer_id  salesman_id
0    False    False    False    False    False
1     True    False    False    False    False
2    False    False     True    False    False
3    False    False    False    False     True
4     True    False    False    False    False
5    False    False    False    False    False
6     True    False    False    False     True
7    False    False    False    False    False
8    False    False    False    False    False
9    False    False    False    False    False
10     True     True    False    False     True
11    False     True     True     True     True

PS C:\Users\Vedvyass M\Music\Nutri-vision>
```

Experiment – 14 :

```
import pandas as pd
```

```
import numpy as np
```

```
data = {  
    "ord_no": [70001, np.nan, 70002, 70004, np.nan, 70005, "...", 70010, 70003,  
70012, np.nan, 70011],  
    "purch_amt": [150.5, 270.65, 65.26, 110.50, 948.50, 200.60, 5760.00,  
1983.43, 2480.40, 250.45],  
    "ord_date": ["?", "2012-09-10", np.nan, "2012-08-17", "2012-09-10",  
"2012-09-09", "2012-09-10", "2012-10-10",  
"2012-06-27", "2012-08-17", "2012-04-25"],  
    "customer_id": [3002, 3001, 3001, 3003, 3002, 3001, 3004, "...", 3002, 3001,  
3001],  
    "salesman_id": [5002, 5003, 5001, np.nan, 5002, 5001, "?", 5003, 5002,  
5003, "..."]  
}
```

```
df = pd.DataFrame(data)
```

```
print("Original DataFrame:")
```

```
print(df)
```

```
df.replace("?", "---", np.nan, inplace=True)
```

```
print("\nDataFrame after replacing invalid values with NaN:")
```

```
print(df)
```

```
df["ord_no"].fillna(df["ord_no"].mean(), inplace=True)
```

```
df["purch_amt"].fillna(df["purch_amt"].mean(), inplace=True)
```

```
df["ord_date"].fillna("NotAvailable", inplace=True)
```

```
df["customer_id"].fillna(df["customer_id"].mode()[0], inplace=True)
```

```
df["salesman_id"].fillna(df["salesman_id"].mode()[0], inplace=True)
```

```
print("\nDataFrame after filling missing values:")
```

```
print(df)
```

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value})' place on the original object.

```
df["ord_date"].fillna("NotAvailable", inplace=True)
```

DataFrame after filling missing values:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
0	70001.0	150.500	2012-10-05	3002.0	5002.0
1	70006.0	270.650	2012-09-10	3001.0	5003.0
2	70002.0	65.260	NotAvailable	3001.0	5001.0
3	70004.0	110.500	2012-08-17	3003.0	5002.0
4	70006.0	948.500	2012-09-10	3002.0	5002.0
5	70005.0	200.600	2012-09-09	3001.0	5001.0
6	70006.0	5760.000	2012-09-10	3004.0	5002.0
7	70010.0	1983.430	2012-10-10	3001.0	5003.0
8	70003.0	2480.400	2012-06-27	3002.0	5002.0
9	70012.0	250.450	2012-08-17	3001.0	5003.0
10	70006.0	1222.029	2012-04-25	3001.0	5002.0
11	70011.0	1222.029	NotAvailable	3001.0	5002.0

PS C:\Users\Vedvyass M\Music\Nutri-vision>